# Venkata Lakshmi Parimala Pasupuleti

AI Engineer · HIPAA · PHI Compliance · Multi-Tenant SaaS · RAG Pipelines · LLM Evaluation · LangGraph · Python · Azure Databricks · Prompt Injection Defense · SOC 2 · NIST

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## PROFESSIONAL SUMMARY

AI Engineer with 8+ years shipping production Python systems in HIPAA-regulated, multi-tenant SaaS environments. At Entrata, owned PHI-compliant data pipelines (100% PHI compliance, zero security incidents) and multi-tenant PostgreSQL schemas for 100K+ housing units. Built LeaseI, an 8-agent LangGraph RAG system with code-enforced compliance boundaries and full audit trail. Built AgentProbe, an open-source LLM evaluation framework with CI-integrated regression harness, 5-metric eval engine, and model routing across OpenAI/Anthropic/Groq/Ollama. AI on FHIR implements HIPAA-compliant RBAC over structured medical records. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). East Coast remote. — AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

## WORK EXPERIENCE

### Blue Horse Digital
Sep 2026 – Present

**AI/ML Engineer (Volunteer) · Virginia, USA**

- Developing ML models and data analysis pipelines; building APIs that surface insights from raw business data into application-facing reporting layers.

### Magizhs Tech Solution
May 2025 – Aug 2025

**AI/ML Engineer Intern**

- Built agentic mission-planning system (LangGraph stateful graph: parse → plan → retrieve → validate → conditional routing) with RAG over FAA Part 107 corpus, validation guardrails, and compliance-grounded rejection reasoning.

### George Mason University
Aug 2024 – Aug 2025

**Research Associate · College of Engineering and Computing**

- Built Python/R prediction pipelines across 190+ countries ($R^2$=0.977); applied Isolation Forest anomaly detection across 3 research pipelines; fine-tuned HuggingFace classification model with PyTorch, deployed inference API on Quart.

### Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **100% PHI compliance on AWS S3 document migration** — Libsodium-encrypted pipelines, zero security incidents across the entire resident data migration; HIPAA-adjacent regulatory requirements enforced at the pipeline layer, not just the access layer.
- **Owned multi-tenant PostgreSQL schemas** for eligibility, financial settlement, and document workflows across 100K+ housing units — per-tenant data isolation, schema-level separation, audit-grade data consistency. Optimized 200+ SQL queries (30% gain) via execution plan analysis, indexing, and stored procedure refactoring.
- **Built RabbitMQ-based async processing** for account state updates across thousands of tenant records — retry logic and idempotency guarantees that eliminated double-processing failures; owned through production incidents and annual compliance cycles.
- **Owned annual HUD income-limit pipeline for 7 years** — few thousand federal rows → ~180,000 domain records powering eligibility, rent, and tax-credit calculations. Zero compliance violations. Every release included baseline validation, reconciliation, and rollback preparation.
- Shipped 20+ full-stack modules (Python, PHP, PostgreSQL, JavaScript) across 120+ Agile sprints; led microservices redesign with Docker — 50% scalability improvement, 40% deployment time reduction; built CI/CD automation pipelines eliminating 80% of manual operational workload.

### Entrata India, Pune
Jul 2017 – Sep 2018

**Associate Software Engineer**

- Built property management modules (Python, Java, PostgreSQL); led AWS S3 cloud migration; redesigned document storage architecture migrating hundreds of thousands of records with full data consistency validation.

## PROJECTS

### Leasel: Regulated AI Investigation Platform
LANGGRAPH · HIPAA-ADJACENT · AUDIT TRAIL

Python · LangGraph (8-agent) · FastAPI · TF-IDF + Semantic RAG · SQLite · OCR | github.com/venkatalakshmiparimala/Lease_complaince_rag

- 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional HITL routing when confidence falls below threshold — core rule enforced in code: LLM extracts and reasons; human or deterministic logic makes every final regulated call.
- Full audit trail on every decision: every answer cited to exact source clause or explicitly flagged as unsupported — every escalation logged with reasoning chain and confidence score. No compliance decision resolved without grounded evidence.
- RAG pipeline: section-aware chunking, hybrid TF-IDF + semantic search, reranking, citation grounding to exact clause — retrieved across lease documents, regulatory corpus, configuration state, and historical incidents.
- Achieved 93.6% precision, 87.9% F1 across 10 compliance rules on 1,240 lease PDFs; pass@1 81.7%, pass@3 88.3% via multi-trial sampling with LLM-as-judge scoring.

### AgentProbe: LLM Evaluation Framework
EVAL FRAMEWORK · CI-INTEGRATED

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI evaluation backend with 5-metric engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression harness with golden datasets that blocks merges on quality drops, model routing across OpenAI/Anthropic/Groq/Ollama, and offline + online evaluation loops.
- Cost controls built in: latency and token efficiency are first-class eval metrics; model router selects cheapest provider meeting latency and quality thresholds. Ollama path enables zero-cost local evaluation.

### AI on FHIR: HIPAA-Compliant Medical Records Platform
HIPAA · PHI

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

- NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings over structured PHI records — enforced HIPAA-compliant RBAC, audit logging, and role-scoped data access; reduced manual clinical retrieval time ~60%.

## EDUCATION

**M.S. Data Analytics Engineering** George Mason University, Virginia — May 2026

GPA: 3.87

**B.Tech Computer Science & Engineering** JNTU Anantapur — 2017

## SKILLS

**Languages:** Python, Java, JavaScript, TypeScript, SQL, PL/SQL, R, Bash, Scala

**AI & Agents:** LangChain, LangGraph, RAG Pipelines, Hybrid Search, Semantic Search, Reranking, Semantic Chunking, Contextual Retrieval, Ingestion Pipelines, Citation Grounding, LLM Evaluation, LLM-as-Judge, Golden Datasets, Regression Harnesses, Offline/Online Evaluation, Hallucination Detection, Model Routing, LLMOps, Agent Observability, Prompt Engineering, Guardrails, Prompt Injection Defense, HITL, Multi-Agent Orchestration, Structured Outputs, Cost Controls, Audit Logging

**Compliance & Security:** HIPAA, PHI Compliance, SOC 2, NIST, Multi-Tenant Data Isolation, RBAC, Libsodium Encryption, Audit Logging, Data Residency, Policy Compliance

**Cloud & Infra:** AWS (S3, EC2, RDS, Glue, Lambda, Bedrock, SageMaker), Azure (Databricks, Data Factory, ADLS, AML, App Service), Docker, Kubernetes, Terraform, Pulumi, CI/CD (GitHub Actions), Prometheus, Grafana, OpenTelemetry

**Data & Pipelines:** PostgreSQL, Snowflake, Elasticsearch, Redis, RabbitMQ, Apache Kafka, Celery, Apache Airflow, ETL, Lakehouse Architecture, Data Governance, Data Lineage, Query Optimization

**ML:** PyTorch, HuggingFace, scikit-learn, XGBoost, Isolation Forest, Anomaly Detection, NumPy, Pandas, spaCy, BERT, Embeddings, Feature Engineering

**APIs & Backend:** FastAPI, REST APIs, Spring Boot, Django, Flask, Microservices, Distributed Systems, Idempotency, API Design